Shared Drives: Permissions

Permissions determine what users can do with files and folders on a shared drive. In Windows, access is controlled by both **Share Permissions** and **NTFS (Security) Permissions**. A user's effective access is the most restrictive combination of the two.

Common Permission Levels

Permission	Description
Read	View files and folders, open documents, and run applications.
Write	Create new files and folders, and save changes to existing files (if allowed).
Modify	Read, write, edit, and delete files and folders.
Full Control	Complete access, including changing permissions and taking ownership.

Share Permissions

Share permissions apply when a folder is accessed over the network.

Common levels include:

- **Read** – View files and folders only.
- **Change** – Read, create, modify, and delete files.
- **Full Control** – Complete access, including managing permissions.

NTFS (Security) Permissions

NTFS permissions apply to files and folders stored on NTFS-formatted drives.

Common permissions include:

- **Read & Execute**
- **List Folder Contents**
- **Read**
- **Write**
- **Modify**
- **Full Control**

How to Check or Change Permissions

1. Right-click the shared folder and select **Properties**.
2. Open the **Sharing** tab to view or change **Share Permissions**.
3. Open the **Security** tab to view or modify **NTFS Permissions**.
4. Select a user or group to see their assigned permissions.
5. Click **Edit** (if authorized) to grant or remove permissions.

Best Practices

- Assign permissions to **security groups** rather than individual users.
- Follow the **principle of least privilege** by granting only the access users need.
- Avoid giving **Full Control** unless it is required.
- Regularly review and remove unnecessary permissions.
- Use descriptive group names (for example, Finance_Read or HR_Modify) to simplify management.

Common Permission Issues

Issue	Possible Cause	Resolution
Access Denied	User lacks required permissions	Verify both Share and NTFS permissions.
Can view but cannot edit files	User has Read-only access	Grant Modify or Write permissions if appropriate.
Permission changes not taking effect	Cached credentials or group membership changes	Sign out and back in, or refresh group membership.
Only some users can access the share	Different group memberships	Confirm the correct users are members of the required security groups.

Understanding and correctly configuring both **Share** and **NTFS** permissions helps ensure users have the appropriate level of access while keeping shared data secure.